

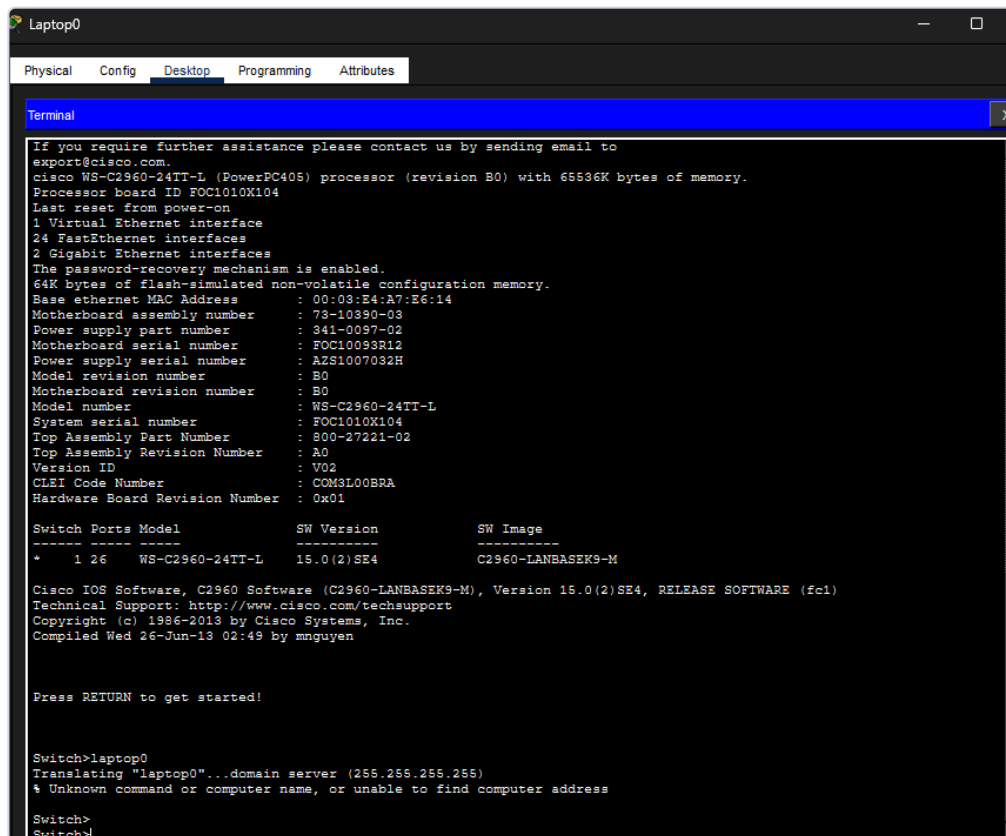
Cisco Packet Tracer - CLI Device Configuration

Comprehensive Engineering Verification Report

Part 1: Connect to the Device Using a Console Connection

Step 1 & 2: Out-of-Band Terminal Emulation

A standard physical roll-over console link was routed between Laptop0 and the Catalyst access switch console port. The serial connection successfully established a raw asynchronous terminal stream at 9600 baud, bypassing standard IP-dependent network transport routes.



```
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:03:E4:A7:E6:14
Motherboard assembly number    : 73-10390-03
Power supply part number      : 341-0097-02
Motherboard serial number     : FOC10093R12
Power supply serial number    : AZS1007032H
Model revision number         : B0
Motherboard revision number   : B0
Model number                  : WS-C2960-24TT-L
System serial number          : FOC1010X104
Top Assembly Part Number      : 800-27221-02
Top Assembly Revision Number  : A0
Version ID                   : V02
CLEI Code Number              : COM3L00BRA
Hardware Board Revision Number : 0x01

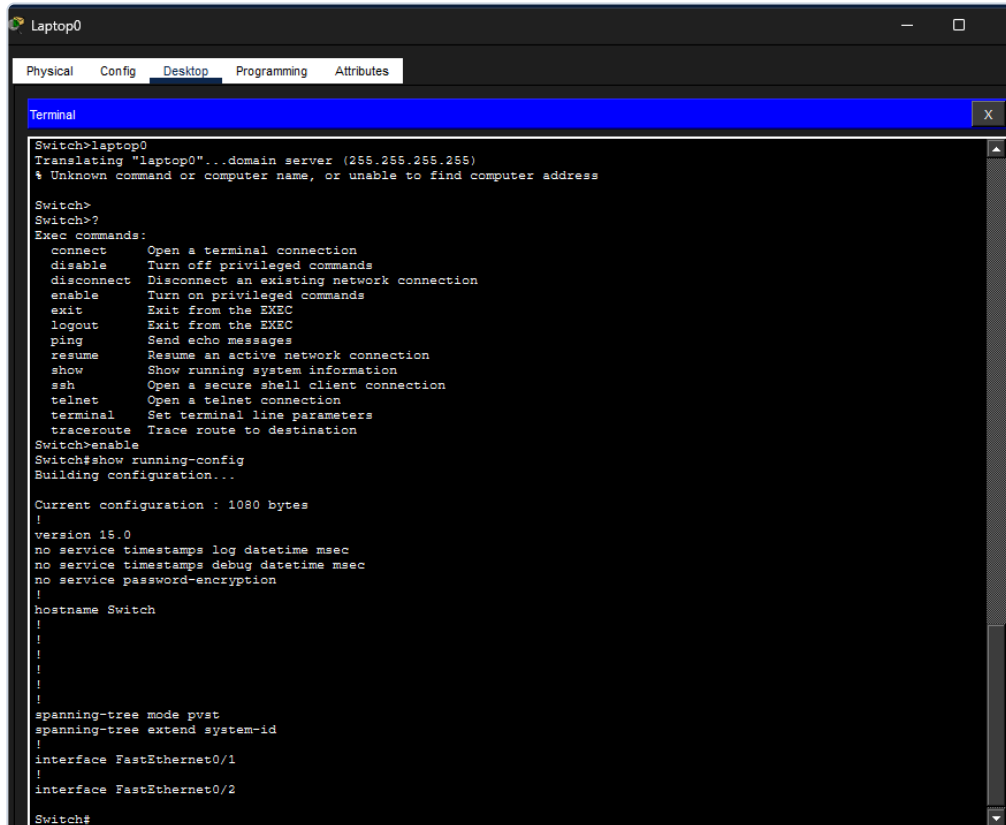
Switch Ports Model          SW Version        SW Image
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*  1 26  WS-C2960-24TT-L  15.0(2)SE4       C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
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Press RETURN to get started!

Switch>laptop0
Translating "laptop0"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch>
Switch>
```



```
Switch>laptop0
Translating "laptop0"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch>
Switch>?
Exit commands:
connect      Open a terminal connection
disable      Turn off privileged commands
disconnect   Disconnect an existing network connection
enable       Turn on privileged commands
exit         Exit from the EXEC
logout       Exit from the EXEC
ping         Send echo messages
resume       Resume an active network connection
show         Show running system information
ssh          Open a secure shell client connection
telnet       Open a telnet connection
terminal     Set terminal line parameters
traceroute   Trace route to destination

Switch>enable
Switch#show running-config
Building configuration...

Current configuration : 1080 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
Switch#
```

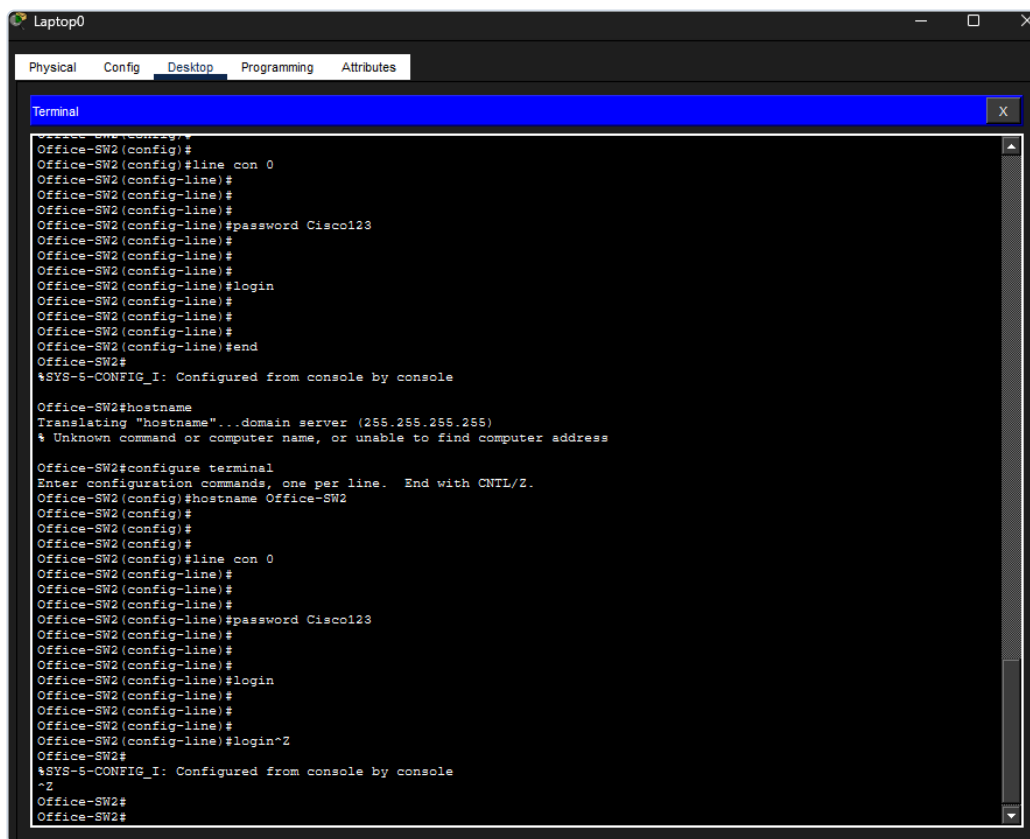
Question: What prompt appears on the switch console screen?

Answer: Switch> (User EXEC Mode)

Part 2: Copy Configuration Information to the Device

Step 1 & 2: Environment Provisioning & Mode Transitions

Advanced system states into Global Configuration mode (configure terminal) to run inline structural scripts. The hostname identifier was initialized as Office-SW2, and AAA access boundaries were applied to the active physical line channel.



```
Laptop0
Physical Config Desktop Programming Attributes
Terminal
Office-SW2(config)#
Office-SW2(config)#line con 0
Office-SW2(config-line)#
Office-SW2(config-line)#
Office-SW2(config-line)#password Cisco123
Office-SW2(config-line)#
Office-SW2(config-line)#login
Office-SW2(config-line)#
Office-SW2(config-line)#
Office-SW2(config-line)#end
Office-SW2#
%SYS-5-CONFIG_I: Configured from console by console

Office-SW2#hostname
Translating "hostname"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Office-SW2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Office-SW2(config)#hostname Office-SW2
Office-SW2(config)#
Office-SW2(config)#
Office-SW2(config)#line con 0
Office-SW2(config-line)#
Office-SW2(config-line)#
Office-SW2(config-line)#password Cisco123
Office-SW2(config-line)#
Office-SW2(config-line)#login
Office-SW2(config-line)#
Office-SW2(config-line)#
Office-SW2(config-line)#login~Z
Office-SW2#
%SYS-5-CONFIG_I: Configured from console by console
^Z
Office-SW2#
Office-SW2#
```

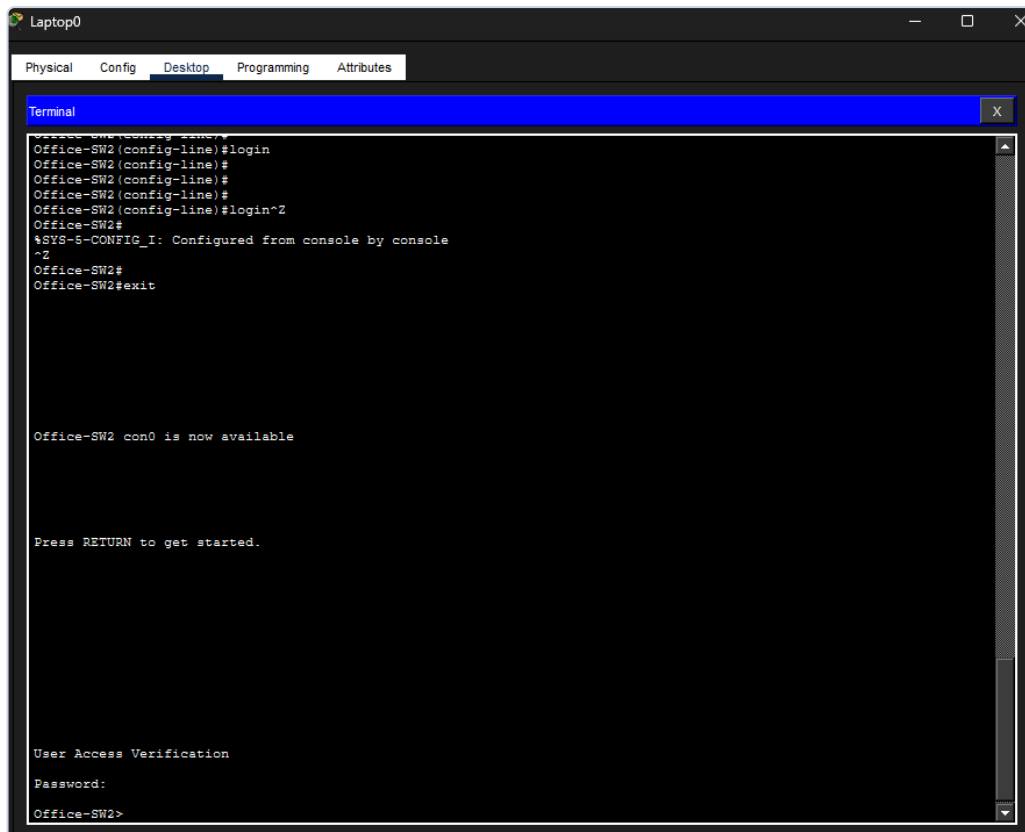
Question: What message is shown on the console after resetting the terminal line?

Answer: User Access Verification followed by an implicit, non-echoing Password: check.

Part 3: Save the Updated Configuration to the Device

Step 1 & 2: Volatile Execution RAM vs. Persistent NVRAM Synchronization

Authenticated into the secure line interface using the credential key. Once administrative privilege scaling was completed via `enable`, current parameters were safely written from volatile active RAM into stable storage cells using the `copy running-config startup-config` tool command sequence.



```
Laptop0
Physical Config Desktop Programming Attributes
Terminal
Office-SW2(config-line)#enable
Office-SW2(config-line)#login
Office-SW2(config-line)#
Office-SW2(config-line)#
Office-SW2(config-line)#login^Z
Office-SW2#
$SYS-5-CONFIG_I: Configured from console by console
^Z
Office-SW2#
Office-SW2#exit

Office-SW2 con0 is now available

Press RETURN to get started.

User Access Verification
Password:
Office-SW2>
```

```
User Access Verification

Password:

Office-SW2> enable
Office-SW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Office-SW2#
```

Part 4: Laboratory Assessment & Completion Summary

Step 3: Verification Logs & Final Engine Score Index

Executed a cold software restart sequence (reload) to test device persistence capabilities. The system completed its bootstrap checks flawlessly, validated security states from NVRAM upon bootup, and officially registered a full performance index.

```
User Access Verification
Password:

Office-SW2> enable
Office-SW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Office-SW2#reload
Proceed with reload? [confirm]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0003.E4A7.E614
Xmodem file system is available.
Initializing Flash...
flashfs(0): 2 files, 0 directories
flashfs(0): 0 orphaned files, 0 orphaned directories
flashfs(0): Total bytes: 64016384
flashfs(0): Bytes used: 4671565
flashfs(0): Bytes available: 59344819
flashfs(0): flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs): installed, fsid: 3
Parameter Block Filesystem (pb): installed, fsid: 4

Loading "flash:/2960-lanbasek9-mz.150-2_SE4.bin"...
***** [OK]
Smart Init is enabled
smart init is sizing iocmem
                                TYPE      MEMORY REQ
TOTAL:                          0x00000000
Rounded IOCMEM up to: 0MB.
Using 6 percent iocmem. [0MB/512Mb]

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Initializing flashfs...
fsck: Disable shadow buffering due to heap fragmentation.
flashfs(2): 2 files, 1 directories
flashfs(2): 0 orphaned files, 0 orphaned directories
flashfs(2): Total bytes: 32514048
flashfs(2): Bytes used: 11362128
flashfs(2): Bytes available: 20561920
flashfs(2): flashfs fsck took 2 seconds.
flashfs(2): Initialization complete....done Initializing flashfs.
Checking for Bootloader upgrade...
Boot loader upgrade not required (Stage 2)
POST: CPU MIC register Tests : Begin
POST: CPU MIC register Tests : End, Status Passed
POST: PortASIC Memory Tests : Begin
POST: PortASIC Memory Tests : End, Status Passed
POST: PortASIC Port Loopback Tests : Begin
POST: PortASIC Port Loopback Tests : End, Status Passed
POST: CPU MIC interface Loopback Tests : Begin
POST: CPU MIC interface Loopback Tests : End, Status Passed
POST: PortASIC RingLoopback Tests : Begin
POST: PortASIC RingLoopback Tests : End, Status Passed
POST: PortASIC CAM Subsystem Tests : Begin
POST: PortASIC CAM Subsystem Tests : End, Status Passed
POST: PortASIC Port Loopback Tests : Begin
POST: PortASIC Port Loopback Tests : End, Status Passed
Waiting for Port download...Complete

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1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
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Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0037-02
Motherboard serial number      : FOC101039R12
Power supply serial number     : A2Z1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COMSLO08RA
Hardware Board Revision Number : 0x01

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*   1 26  WS-C2960-24TT-L      15.0(2)SE4  C2960-LANBASEK9-M

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```

```
Terminal

Continuing the boot-loader upgrade...
Boot loader upgrade not required (Stage 2)
POST: CPU MIC register Tests : Begin
POST: CPU MIC register Tests : End, Status Passed
POST: PortASIC Memory Tests : Begin
POST: PortASIC Memory Tests : End, Status Passed
POST: PortASIC Port Loopback Tests : Begin
POST: PortASIC Port Loopback Tests : End, Status Passed
POST: CPU MIC interface Loopback Tests : Begin
POST: CPU MIC interface Loopback Tests : End, Status Passed
POST: PortASIC RingLoopback Tests : Begin
POST: PortASIC RingLoopback Tests : End, Status Passed
POST: PortASIC CAM Subsystem Tests : Begin
POST: PortASIC CAM Subsystem Tests : End, Status Passed
POST: PortASIC Port Loopback Tests : Begin
POST: PortASIC Port Loopback Tests : End, Status Passed
Waiting for Port download...Complete

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http://www.cisco.com/wd/export/crypto/cool/stg.html
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Top Assembly Revision Number   : A0
Version ID                     : V02
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Press RETURN to get started!

User Access Verification
Password:
```

Congratulations Guest! You completed the activity.

Overall Feedback [Assessment Items](#) Connectivity Tests

Expand/Collapse All Show Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedback
- Network				
- Office-SW2				
- Console Line				
✓ Login	Correct	1	Physical	
✓ Password	Correct	1	Other	
✓ Host Name	Correct	1	Other	
✓ Startup Config	Correct	1	Other	

Final Summary: The Cisco Packet Tracer assessment engine confirms a 100% completion profile (Points: 6/6). All criteria items regarding Hostname assignment, Console Line settings, Password variables, and configuration replication checks have been successfully satisfied.